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"""
Project 2: Expense Tracker
DecodeLabs Python Industrial Training Kit — Batch 2026

Continuously accepts expense amounts from the user, accumulates them
into a running total (the "accumulator pattern"), and reports the
final amount spent once the user enters the sentinel command "quit".
"""


def expense_tracker():
    total = 0.0           # Accumulator — initialized OUTSIDE the loop
    transaction_count = 0

    print("=== Expense Tracker ===")
    print("Enter an expense amount and press Enter.")
    print("Type 'quit' when you're done.\n")

    while True:
        raw_value = input("Expense: ").strip()

        # --- Kill switch: sentinel value
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halts the loop ---
        if raw_value.lower() == "quit":
            break

        # --- Gatekeeper: defensive type
conversion ---
        try:
            new_expense = float(raw_value)
        except ValueError:
            print("Invalid data — please
enter a number.\n")
            continue

        if new_expense < 0:
            print("Expenses can't be
negative — try again.\n")
            continue

        # --- Accumulator pattern:
State(new) = State(old) + Input ---
        total = total + new_expense
        transaction_count += 1
        print(f"  Added ${new_expense:.2f}
-> Running total: ${total:.2f}\n")

    # --- Output phase: display logic kept
separate from the calculation above ---
    print("\n=== Summary ===")
    print(f"Transactions recorded:
{transaction_count}")
    print(f"Total Spent: ${total:.2f}")
```

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if __name__ == "__main__":
    expense_tracker()
```